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Cite this: *Analyst*, 2013, **138**, 1504

Enhancement of the spectral selectivity of complex samples by measuring them in a frozen state at low temperatures in order to improve accuracy for quantitative analysis. Part I. Raman spectroscopic compositional analysis of synthetic hydrocarbon mixtures†

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A simple strategy for enhancing the Raman spectral selectivity of complex mixture samples by measuring them in a frozen state at low temperatures has been demonstrated and proven to improve the accuracy for compositional analysis. For evaluation, the Raman spectra of synthetic hydrocarbon mixtures that were composed of eleven hydrocarbons (*n*-hexane, *n*-heptane, *n*-octane, *n*-nonane, isooctane, cyclohexane, methylcyclohexane, benzene, toluene, xylene, and indan) were continuously collected during the elevation of their temperature from cryogenic to near room temperature. The accuracy of determination of *n*-paraffin concentrations improved significantly when the samples were measured at the temperature range between approximately −175 and −155 °C in comparison to the measurements at room temperature. However, the improvement of accuracy was relatively marginal for the concentration determination of naphthenic and aromatic components. Since *n*-paraffins are easily compressible and deformable in frozen conditions, the subsequent spectral variations could be diverse depending on their molecular structures. Due to this fact, the spectral discrimination among the paraffin components, as well as in comparison to other components, was enhanced and the improved spectral selectivity eventually led to more accurate determination of concentrations. Overall, the proposed strategy is simple and effective, so it is applicable for analysis of real complex mixture samples.

Received 29th October 2012
Accepted 4th January 2013

DOI: 10.1039/c2an36575h

www.rsc.org/analyst

Introduction

When a simple and fast compositional analysis of complex samples without the use of separation steps is in great demand, vibrational spectroscopic methods including near-infrared (NIR)[1–3] and Raman spectroscopy[4–7] are frequently employed. In this situation, the spectra are directly collected from raw complex samples without any further treatments, so that the resulting spectral features overlap extensively with the degraded compositional specificity. Therefore, the use of multivariate methods[8,9] is necessary in order to empirically correlate the spectral features with the corresponding concentrations or properties of the samples. When an analyst would like to improve the measurement accuracy in this situation, only limited approaches are available such as acquisition of reproducible spectra with a high signal-to-noise ratio.

The enhancement of the spectral selectivity among mixture components could be a straightforward way to establish a more accurate correlation with the variation in the concentrations or properties. When a complex mixture sample is measured in a frozen state at a low temperature, the molecular vibrations of the individual components become simpler. Therefore, the resulting spectral bandwidths of the components become narrower and the overall spectral feature could be simplified with less spectral overlapping. In addition, newly emerging bands would become available in these conditions due to the denser intermolecular space that freezes molecular vibration as well as rotation.[10–13] Therefore, the spectral acquisition of samples at frozen conditions could possibly improve the spectral selectivity.

We studied a strategy to enhance the spectral selectivity by measuring complex mixture samples in a frozen condition (much below their freezing point). In part I, the effectiveness of the strategy was primarily investigated for the quantitative compositional analysis of synthetic hydrocarbon mixtures composed of eleven components (*n*-hexane, *n*-heptane, *n*-octane, *n*-nonane, isooctane, cyclohexane, methylcyclohexane, benzene, toluene, xylene, and indan) using Raman spectroscopy. These

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† For part II see ref. 25.

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components were selected in order to emulate a practical sample, naphtha (a mixture of C_4 through C_9 hydrocarbons), as well as to evaluate the strategy when the sample spectra were highly overlapping due to similar molecular structures. The samples were initially frozen in liquid nitrogen and simply exposed to ambient conditions in order to continuously increase the sample temperature. During the temperature elevation, Raman spectra were continuously collected at a regular interval and a total of 90 spectra datasets were obtained for each sample.

Principal component analysis (PCA)[8,9] was used in order to observe the overall spectral variation of all of the samples occurring during the temperature elevation from cryogenic to near room temperature, and the resulting score variation in the two dimensional domain was thoroughly investigated. Then, partial least squares (PLS)[8,9] was performed to determine the concentrations of each component, the spectral datasets collected at each period and the resulting accuracies were compared. The optimal temperatures that resulted in improved accuracy for each component were identified and the relevant justification for the improvement was searched by examining the varied spectral features of the components in the corresponding conditions.

Experimental section

Samples and Raman spectral collection

The eleven pure hydrocarbons used in this study were purchased from Sigma-Aldrich. A list of the 11 components and their melting temperatures is shown in Table 1. The mixture samples were prepared by mixing the appropriate amounts of each component according to the random experimental design. The concentration maxima and minima for each component are also summarized in the table. A total of 25 mixture samples were prepared. For a reliable multivariate calibration, correlations among the concentrations of the individual components should not be present. The concentration correlations among the components were checked by examining the R^2 values calculated by plotting the concentration of one component *versus* that of other components. In all cases, the R^2 values were below 0.36 and no noticeable correlations were observed.

Table 1 A list of the 11 hydrocarbon components in the mixture samples and the concentration ranges of each component in the sample dataset. The melting temperatures of each component are also shown

Number of carbon atoms	Component	Type	Mp (°C)	Max. (wt%)	Min. (wt%)
6	*n*-Hexane	*n*-Paraffin	−95	18.72	5.57
	Cyclohexane	Naphthene	6.5	18.46	5.43
	Benzene	Aromatic	5.5	16.30	5.28
7	*n*-Heptane	*n*-Paraffin	−91	17.52	6.19
	Methylcyclohexane	Naphthene	−126	5.14	0.46
	Toluene	Aromatic	−93	17.41	6.32
8	*n*-Octane	*n*-Paraffin	−57	18.67	6.90
	Isooctane	*i*-Paraffin	−107	5.58	0.32
	Xylene	Aromatic	12.5	15.55	5.56
9	*n*-Nonane	*n*-Paraffin	−53	19.00	8.54
	Indan	Aromatic	−51	5.58	0.61

The Raman spectra were collected using a wide area illumination (WAI) scheme (PhAT System, Kaiser Optical Inc., Ann Arbor, MI, USA) as has been used in previous studies.[14–16] Laser (785 nm) was illuminated in a circular fashion with a diameter of 6 mm to cover a large sample area and the laser power at the sample was 200 mW approximately. A glass vial (diameter: 26 mm) containing a sample (20 mL) was initially immersed in liquid nitrogen for 10 minutes. Then, the vial containing the sample was quickly taken out of the liquid nitrogen and immediately positioned at the focal point of the WAI scheme (focal length: 250 mm approximately) for the spectral collection. As expected from the pictures in Fig. 1, the samples were bi-phasic of both solid and liquid states

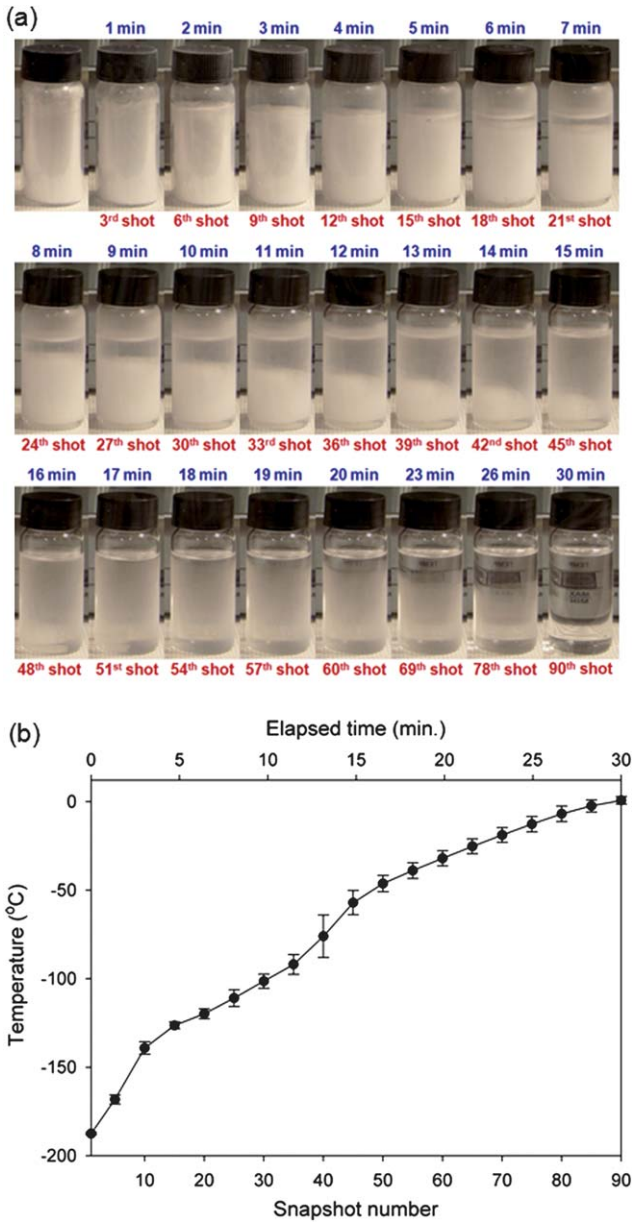

Fig. 1 Pictures of one selected sample contained in a glass vial during the increase in temperature from cryogenic to near room temperature (a) and the actual temperature variation of the samples according to the snapshot number and elapsed time (b).

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approximately in the 20th–50th snapshot range; it was required to illuminate laser radiation on the same location on a sample for reproducible spectral collection. Therefore, for all cases, the same amount of samples was filled into a glass vial and laser radiation was reproducibly illuminated on the middle part of the samples. The sample vial and the WAI probe were located inside a custom-made N_2-purged plastic box in order to prevent extensive formation of frost on the wall of the glass vial. The duration for each snapshot lasted 20 seconds, resulting in a total of 90 snapshots (spectra) for each sample.

Fig. 1(a) shows the actual pictures of one selected sample in a glass vial during the temperature elevation from cryogenic to near room temperature. The corresponding snapshot number and elapsed time are indicated at the bottom and top of each picture, respectively. As soon as the sample vial was removed from the liquid nitrogen, frost immediately started to form on the wall, but it became thinner at 3 minutes. The frozen sample began to thaw at 5 minutes (approximately −126 °C) and turned completely into a liquid phase at 18 minutes (approximately −39 °C). The Raman spectral feature of frost (basically water) was very weak and so its influence on the overall spectral feature of the sample was not significant.

Fig. 1(b) shows the actual temperature variation of the samples during the spectral collection, in which the *x*-axis is expressed as both the snapshot number and the elapsed time. The error bars at each snapshot correspond to the standard deviations of the measured temperatures from all 25 samples. Since the melting temperatures of each component varied greatly as shown in Table 1, the pattern of the temperature increase should be different from sample to sample. The samples mostly started to melt between approximately −120 and −80 °C, so there were larger temperature variations (larger magnitude of error bar) among the samples in this temperature range.

Triplicate spectra were collected for each sample. Each sample was divided into three different glass vials and then the Raman spectra of these divided samples were collected separately by varying the temperature as described above. The resolution of the spectra was 4 cm^{-1}. All spectral pre-processing and multivariate methods, including baseline correction, normalization, PCA and PLS, were performed using MATLAB Version 7.0 (Math Works Inc., MA, USA).

Results and discussion

Spectral variations of pure components during temperature elevation

For a simple study, *n*-hexane, cyclohexane, and benzene were initially selected as the representative components for *n*-paraffin, naphthene, and aromatic hydrocarbon, respectively, and their spectral variations from cryogenic to near room temperature were examined. Fig. 2(a) shows the Raman spectra of *n*-hexane collected during the increase in temperature between the 1230 and 190 cm^{-1} range. The baselines of the

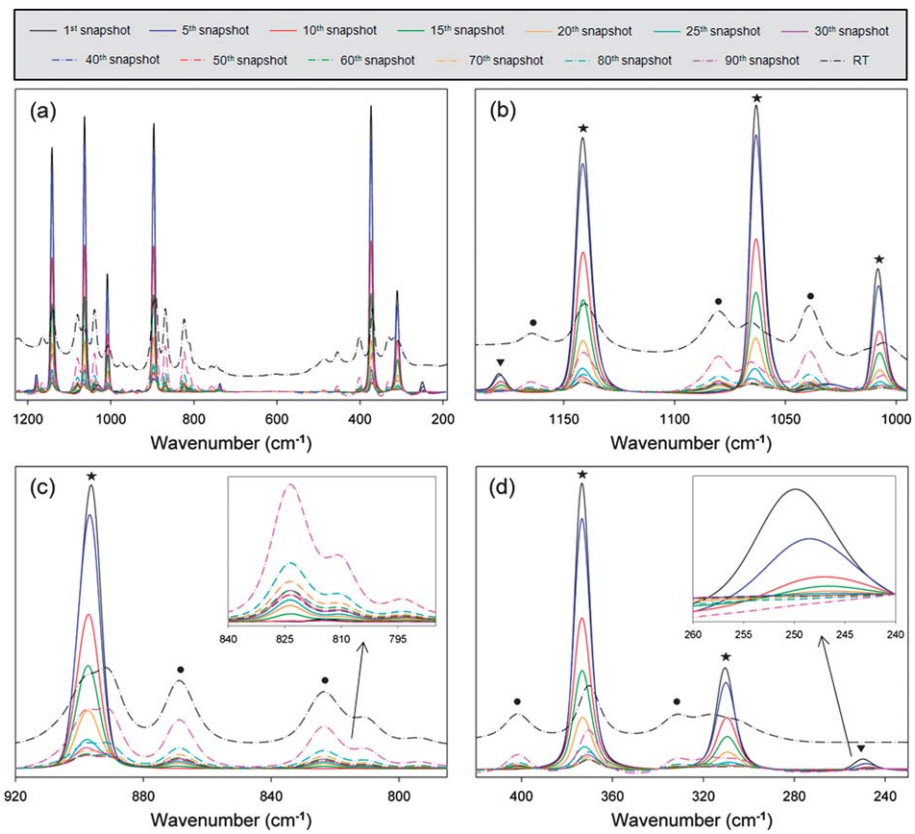

Fig. 2 Baseline-corrected Raman spectra of *n*-hexane collected during the temperature elevation from cryogenic to near room temperature in the 1230–190 cm^{-1} (a), 1190–995 cm^{-1} (b), 920–785 cm^{-1} (c) and 420–230 cm^{-1} (d) ranges.

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spectra were linearly corrected at nine different wavenumbers (1230, 1120, 945, 660, 565, 480, 340, 240, and 190 cm^{-1}). In Fig. 2(a), only 13 spectra corresponding to the 1st, 5th, 10th, 15th, 20th, 25th, 30th, 40th, 50th, 60th, 70th, 80th and 90th snapshots are displayed for clarity. The spectrum of *n*-hexane collected at room temperature (black dashed line) is shown and intentionally offset for easier comparison. For more detailed examination, the spectral features in the 1190–995, 920–785 and 420–230 cm^{-1} ranges are highlighted in Fig. 2(b)–(d), respectively. As shown in the figures, the Raman intensities of the major bands are strongest at the 1st snapshot and gradually decrease as the temperature increases. The intensity variation is much more substantial during the earlier snapshots, while, it is relatively minimal after the 30th snapshot.

Three different trends were observed in the spectral variation shown in Fig. 2. First, the Raman bands are present in both the 1st snapshot and the room temperature spectrum, but they are very intense at low temperatures including the bands at 1142, 1063, 1008, 898, 373 and 310 cm^{-1} (marked as stars). Second, the Raman bands are only visible at near cryogenic conditions such as the bands at 1179 and 250 cm^{-1} (marked as triangles). Third, the Raman bands appearing only at room temperature are identified as the bands at 1164, 1080, 1039, 869, 824, 402 and 331 cm^{-1} (marked as circles).

The intensities of the bands marked by the stars are strongest in the 1st snapshot. These bands correspond to the vibration of the C–C backbone or skeleton of the hydrocarbon.[17–20] At cryogenic temperatures, the hydrocarbon backbone became highly rigid and the arrangement of the *n*-hexane was being condensed; therefore, an increase in the Raman intensities occurred. As the temperature increased, the rigid structure and dense arrangement of the molecules relaxed and the Raman intensities decreased as a result.

The weak bands at 1179 and 250 cm^{-1} bands (marked as triangles) appear only in the early snapshots and they disappear near room temperature. The appearance of these bands at low temperatures is difficult to explain clearly. However, it is presumably a result of the chain expansion–contraction motion of the linear hydrocarbons when they are completely frozen.[21–23] The bands marked by circles are only present when the *n*-hexane returned to a liquid state. These bands correspond to the vibrations of $-CH_2$ and $-CH_3$.[17–20]

Fig. 3(a) shows the Raman spectra of cyclohexane collected at different snapshots in the 1200–330 cm^{-1} range. The spectral features in the 1170–1010, 815–775 and 440–365 cm^{-1} ranges are highlighted separately in Fig. 3(b)–(d), respectively. The spectrum of cyclohexane collected at room temperature (black dashed line) is also shown. Similarly, the intensities of most of the Raman bands decreased with an increase in the temperature, while the intensity of the band at 801 cm^{-1} (at room temperature) was nearly unchanged as shown in Fig. 3(c). The position of this band shifted to a higher wavenumber at

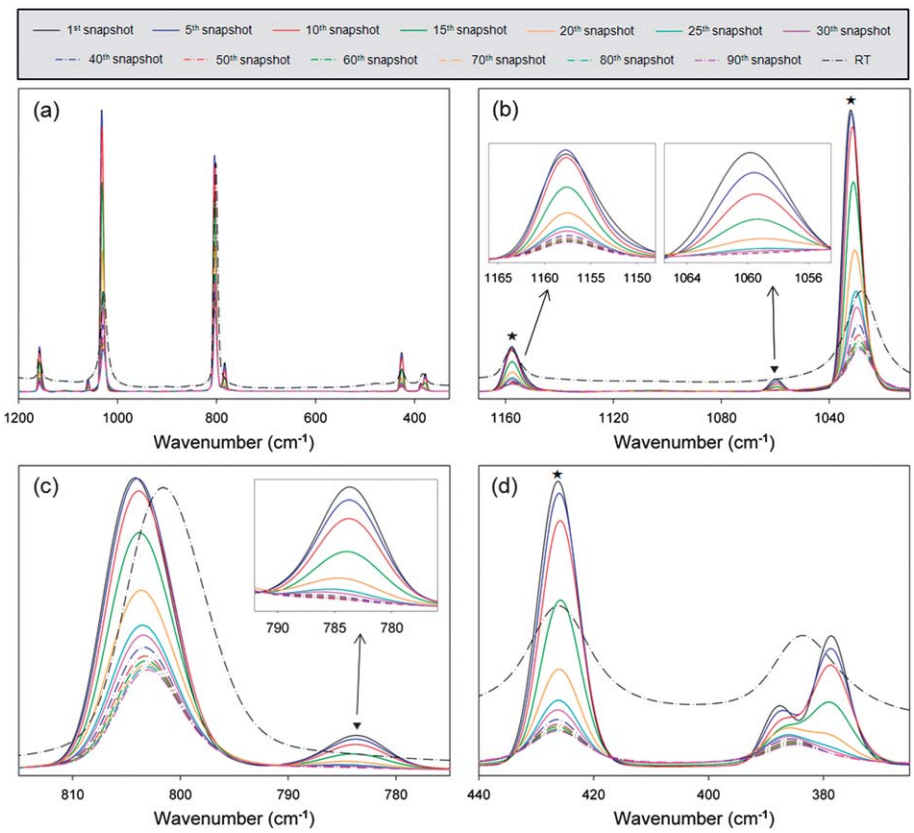

Fig. 3 Baseline-corrected Raman spectra of cyclohexane collected during the temperature elevation from cryogenic to near room temperature in the 1200–330 cm^{-1} (a), 1170–1010 cm^{-1} (b), 815–775 cm^{-1} (c) and 440–365 cm^{-1} (d) ranges.

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cryogenic temperatures and its intensity continuously decreased with an increase in temperature. The bands at 1158, 1032, and 426 cm^{-1} (marked as stars) were observable in both the 1st snapshot and the room temperature spectrum, while their intensity was higher at low temperatures. The bands at 1060 and 784 cm^{-1} (marked as triangles) were only present in cryogenic conditions. The 383 cm^{-1} band observed at room temperature split into two peaks (388 and 379 cm^{-1}) in the earlier snapshots. Overall, the shift and merge of the cyclohexane bands were more obvious than the spectral variation of *n*-hexane during the increase in the temperature.

Fig. 4(a) shows the Raman spectra of benzene collected at different snapshots in the 1230–565 cm^{-1} range. The spectral features in the 1190–1160, 1020–975, and 890–590 cm^{-1} ranges are shown individually in Fig. 4(b)–(d), respectively. The 1174 cm^{-1} band seems to have two overlapped features in cryogenic conditions, while these merged into a single band with an increase in temperature as shown in Fig. 4(b). The position of this band shifted slightly to a higher wavenumber at room temperature. The 993 cm^{-1} band at room temperature shifted to 990 cm^{-1} at cryogenic temperatures (see Fig. 4(c)). The intensity of the 990 cm^{-1} band was not changed significantly even in cryogenic conditions and then it gradually decreased with increase in temperature. The 1012, 877, 865 and 854 cm^{-1} bands at cryogenic temperatures showed similar behavior upon increase in temperature. The intensities of these bands decreased and their bands simultaneously shifted to a lower wavenumber. As shown in Fig. 4(d), the position of the 606 cm^{-1} band remained relatively unchanged, while its intensity decreased with variation in temperature.

Fig. 5 shows the Raman spectra of all of the paraffins (*n*-hexane, *n*-heptane, *n*-octane, *n*-nonane, and isooctane) collected in the 1st snapshot (left column) and at room temperature (right column). The spectral bands in the 1230–990 cm^{-1} (top), 990–725 cm^{-1} (middle) and 550–190 cm^{-1} (bottom) ranges are highlighted separately. The overall spectral features of the paraffins in the 1st snapshot and at room temperature are largely different as was expected. The peaks are simpler and more distinct at cryogenic temperatures, so the degree of spectral overlapping among these paraffin compounds was greatly reduced, especially in the 550–190 cm^{-1} range.

When aliphatic hydrocarbons were frozen at temperatures much lower than their melting temperatures, the vibrations of the side chains (including $-CH_2$ rocking, wagging, and twisting; $-CH_3$ terminal rocking; and in-plane $-CH_3$ rocking) were significantly suppressed.[17–20] This suppression resulted in the simpler vibrational peaks shown above. In addition, it is known that the longitudinal acoustic vibrations become significant when these molecules are frozen at low temperatures.[21–23] These vibration modes correspond to the skeletal and lattice vibrations of the backbones and they vary depending on the chain length. There are two kinds of longitudinal acoustic

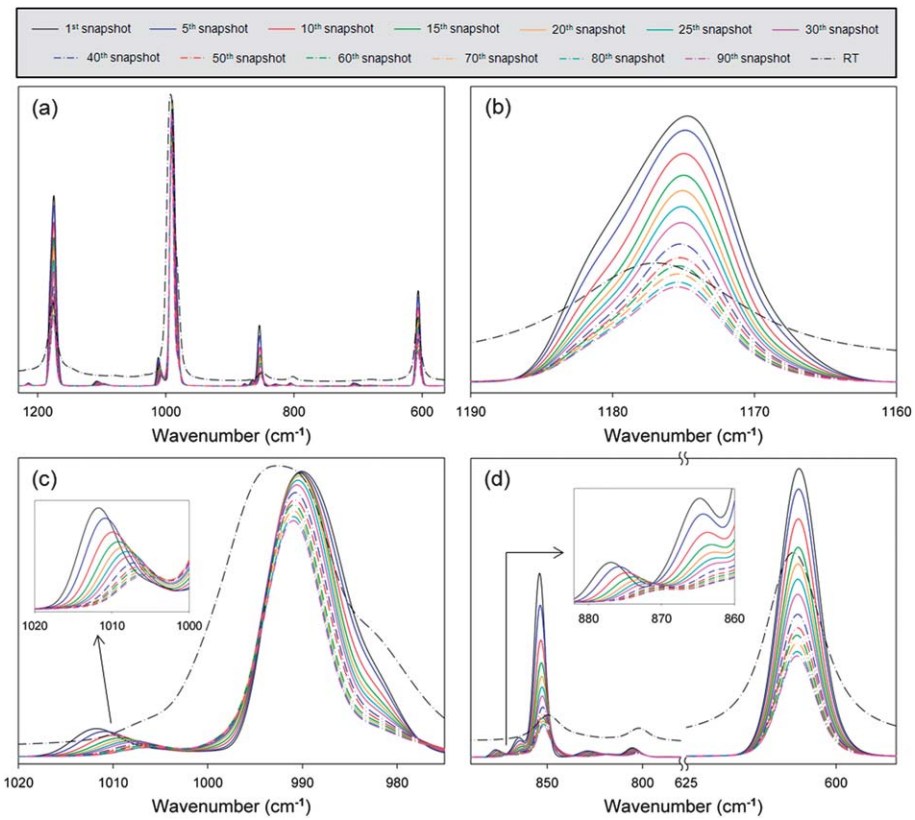

Fig. 4 Baseline-corrected Raman spectra of benzene collected during the temperature elevation from cryogenic to near room temperature in the 1230–565 cm^{-1} (a), 1190–1160 cm^{-1} (b), 1020–975 cm^{-1} (c) and 890–590 cm^{-1} (d) ranges.

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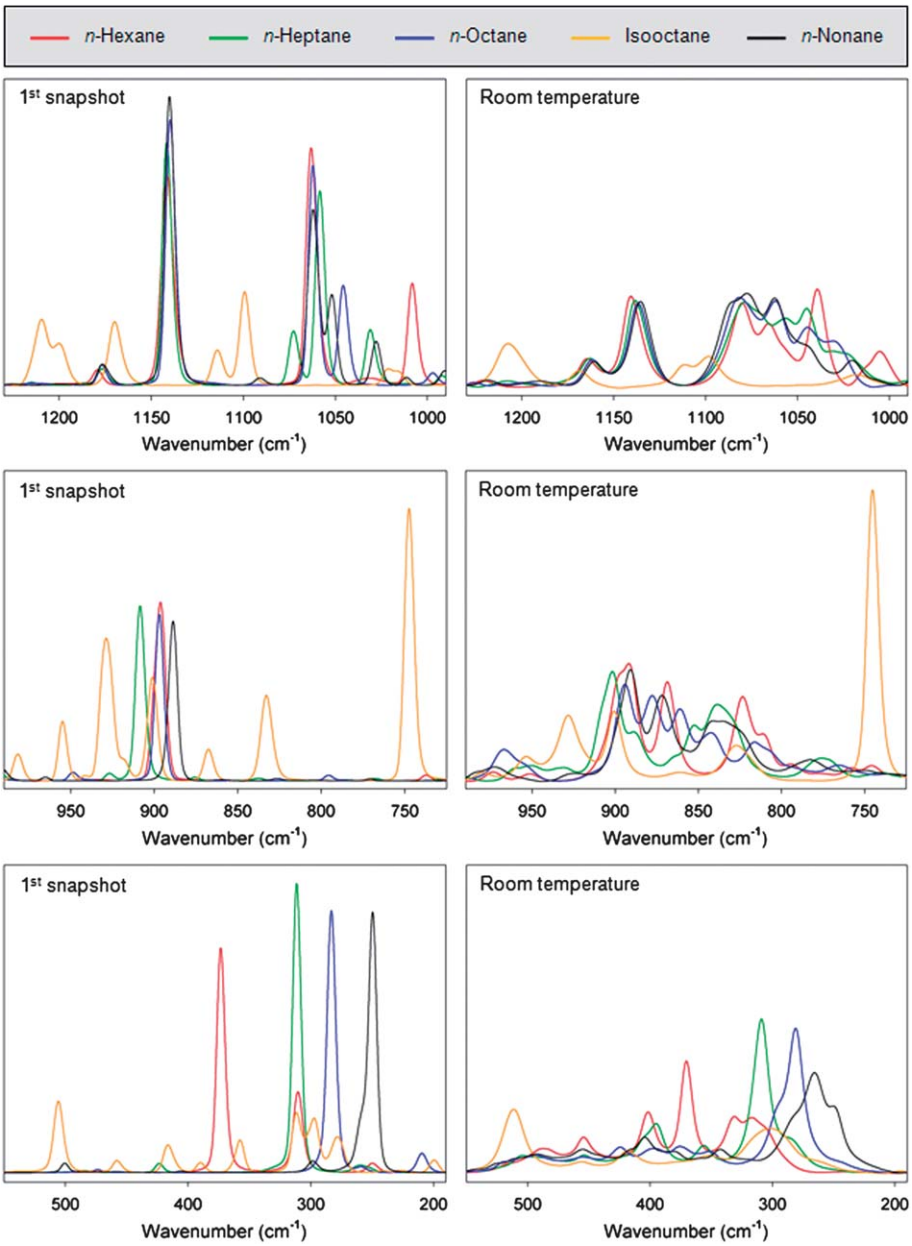

Fig. 5 Normalized Raman spectra of paraffin hydrocarbons collected in the 1st snapshot (left column) and at room temperature (right column) in the 1230–990 cm^{-1} (top), 990–725 cm^{-1} (middle) and 550–190 cm^{-1} (bottom) ranges.

vibrations in a frozen state.[21–23] The first type is the combination of in-plane $-CH_3$ rocking, C–C stretching, and C–C–C deformation. The corresponding band appears in the 920–880 cm^{-1} range and the position varies depending on the chain length of the linear hydrocarbons (*n*-hexane: 898 cm^{-1}, *n*-heptane: 905 cm^{-1}, *n*-octane: 899 cm^{-1} and *n*-nonane: 888 cm^{-1}). The second type is the chain expansion–contraction motion of the C–C backbone. The corresponding band shifts to a lower wavenumber as the chain length increases (*n*-hexane: 373 cm^{-1}, *n*-heptane: 311 cm^{-1}, *n*-octane: 282 cm^{-1} and *n*-nonane: 249 cm^{-1}). The bands at 898 and 373 cm^{-1} shown in Fig. 2(c) and (d) correspond to the two different longitudinal acoustic vibrations of *n*-hexane. Overall, the spectral differentiation among *n*-paraffins was enhanced by performing measurements in a frozen condition due to the induced diverse structural deformations including the longitudinal acoustical vibrations.

Fig. 6 shows the Raman spectra of two naphthenic components (cyclohexane and methylcyclohexane) collected in the 1st snapshot (left column) and at room temperature (right column). The spectral features are highlighted in the 1230–1000 cm^{-1} (top), 1000–725 cm^{-1} (middle), and 580–190 cm^{-1} (bottom) ranges. The spectral features of the naphthenic components at cryogenic and room temperatures are also different in the 580–190 cm^{-1} range. However, the overall difference is not as dramatic as that observed in the spectra of the *n*-paraffins. The structural deformation and/or compression of naphthene molecules would not be facile like paraffin hydrocarbons, since naphthenes possess saturated rings in their structure.

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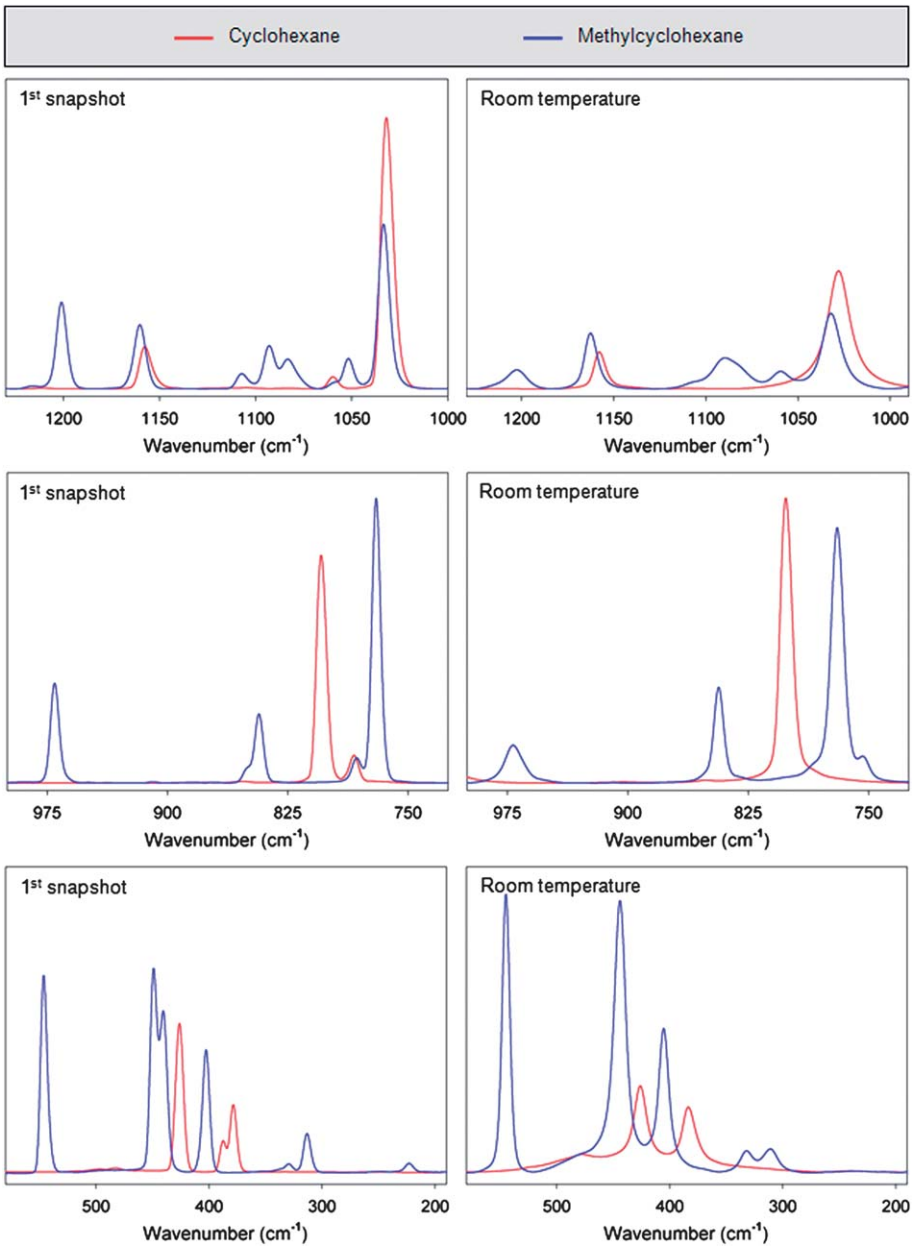

Fig. 6 Normalized Raman spectra of naphthenic hydrocarbons collected in the 1st snapshot (left column) and at room temperature (right column) in the 1230–1000 cm^{-1} (top), 1000–725 cm^{-1} (middle) and 580–190 cm^{-1} (bottom) ranges.

Fig. 7 shows the Raman spectra of four aromatic components (benzene, toluene, xylene and indan) collected in the 1st snapshot (left column) and at room temperature (right column). The spectral features are detailed in the 1230–950 cm^{-1} (top), 950–550 cm^{-1} (middle) and 550–190 cm^{-1} (bottom) ranges. There was not much difference between the spectra collected in the two different conditions, which is similar to the results for the naphthenic components. Due to the presence of a benzene ring in the structures of the aromatic components, the deformability of the structure would not be significant even in a frozen state. The minor spectral variations that occurred at the cryogenic temperatures could be more attributed to the varied vibration of the substituents attached to the benzene ring.

Fig. 8 shows the Raman spectra of a selected sample that was collected in the 1st snapshot and at room temperature. The spectral features in the 1230–700 cm^{-1} (a) and 700–190 cm^{-1} (b) ranges are shown. In the case of the bands marked by stars, their band intensity is higher and the bandwidth is narrower in the cryogenic conditions. The bands indicated by triangles have relatively similar intensities in both conditions, while the position of the band or bandwidth is different. The bands indicated by circles provide a relatively unique feature in cryogenic conditions. Newly emerging bands or better-resolved bands observed in the spectra of the *n*-paraffins in a cryogenic condition are not recognizable in the mixture sample spectrum due to the considerable spectral overlapping with the bands

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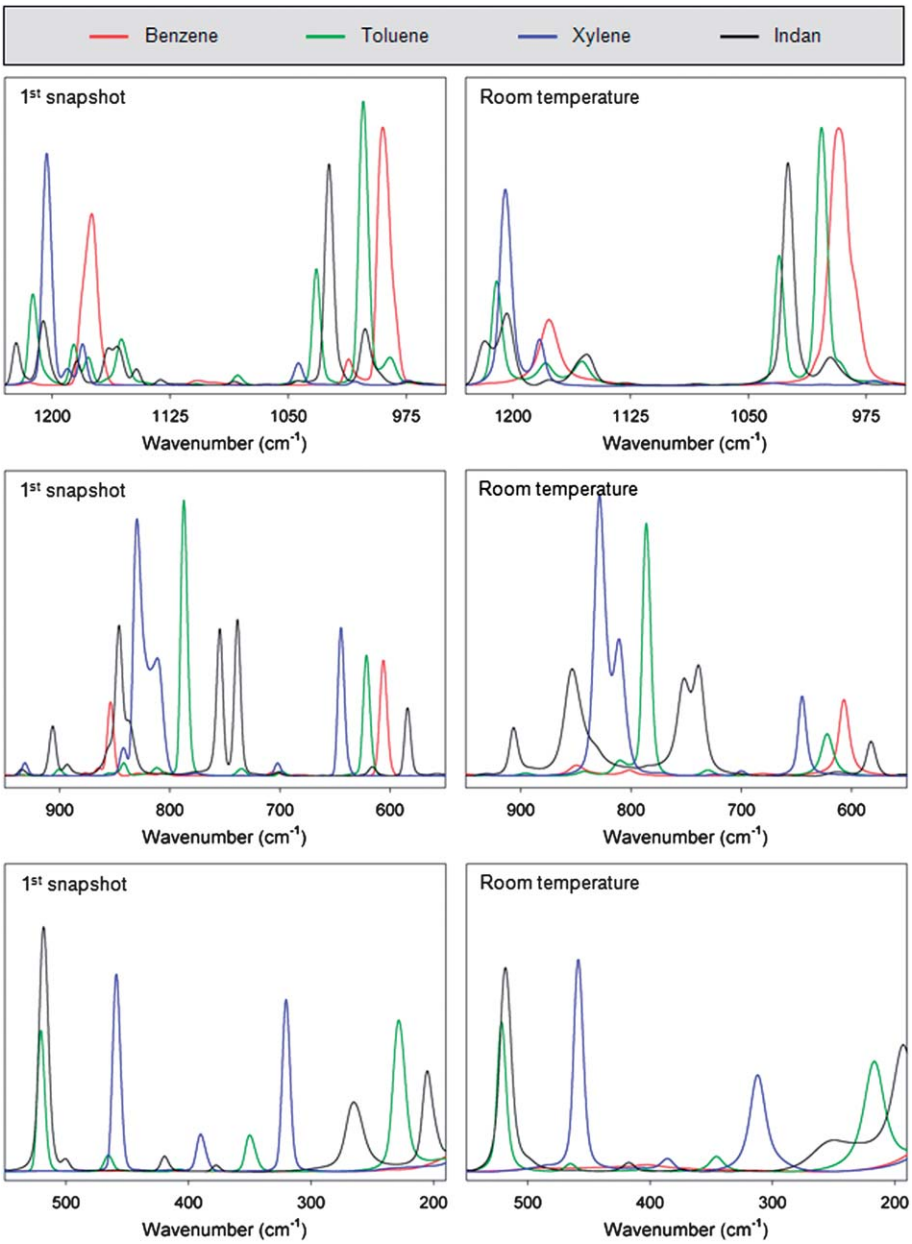

Fig. 7 Normalized Raman spectra of aromatic hydrocarbons collected in the 1st snapshot (left column) and at room temperature (right column) in the 1230–950 cm^{-1} (top), 950–550 cm^{-1} (middle) and 550–190 cm^{-1} (bottom) ranges.

from the other components. However, valuable spectral variations that are able to enhance the spectral selectivity should be available in the spectra and could potentially be effective for improving the accuracy of the quantitative compositional analysis.

Principal component analysis

In order to simultaneously visualize the overall spectral variation in all of the samples from the 1st to the 90th snapshot, PCA was performed using the 1230–190 cm^{-1} range and the variation of the resulting scores was examined. All of the spectra were normalized before performing PCA. For normalization, the baselines of raw spectra were linearly corrected at 1230, 1120, 945, 660, 565, 480, 340, 240 and 190 cm^{-1}, the area under the 1230–190 cm^{-1} range was calculated, and each baseline-corrected spectrum was divided by the corresponding area. Along with the sample spectra collected during the increase in temperature, the spectra collected at room temperature were also included for the PCA.

The score scatter plot constructed using the first and second scores, representing the most significant spectral variation in the dataset, is shown in Fig. 9. For easier comparison, the scores corresponding to the measurements in the 1st (blue), 10th (yellow), 20th (green), 30th (red), 40th (cyan) and 50th snapshots (purple), as well as at room temperature (gray triangle) were only compared. In the earlier stages including the 1st through 20th snapshots, the distribution of the scores is widely spread. This

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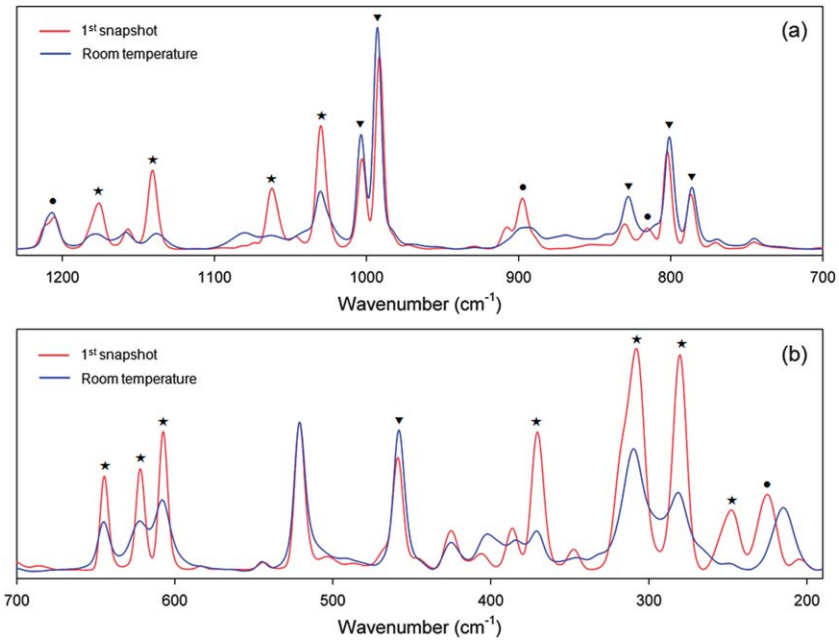

Fig. 8 Normalized Raman spectra of a selected sample collected in the 1st snapshot and at room temperature in the 1230–700 cm^{-1} (a) and 700–190 cm^{-1} (b) ranges.

result indicates the occurrence of large and diverse spectral variations among the samples during this period. The variation of the scores in the 1st snapshot is greatest in both the x (first score) and y (second score) directions. The distribution of the scores starts shrinking in the 30th snapshot and the scores begin to aggregate each other in the 40th snapshot. After the 40th snapshot, the score distribution is nearly identical to that of the room temperature measurement. The larger spectral variation in the earlier snapshots could be advantageous for improving the spectral selectivity among the mixture components, thereby providing more descriptive information for the compositional variation of the individual components.

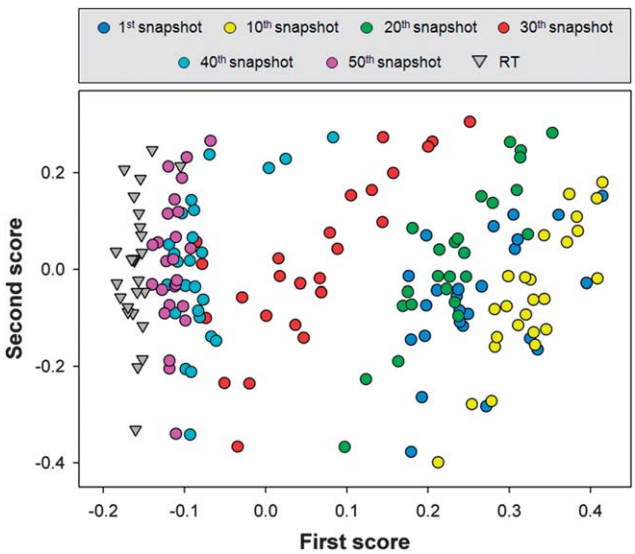

Fig. 9 Score scatter plot (the first *vs.* second scores) showing the variation in the entire sample spectra collected in the 1st through 90th snapshots as well as at room temperature. For easier comparison, the scores corresponding to the measurements in the 1st, 10th, 20th, 30th, 40th and 50th snapshots, as well as at room temperature (gray triangle) are shown.

Determination of the concentrations of individual components in each snapshot

PLS was used to determine the concentrations of each component from the datasets collected in each snapshot and at room temperature. Out of a total of 25 samples (75 spectra), 18 samples (54 spectra) and the remaining 7 samples (21 spectra) were used as the calibration and validation sets, respectively. The samples were randomly assigned into the calibration and validation sets, and this procedure was repeated 200 times for cross-validation. Normalized spectra of the samples in the 1230–190 cm^{-1} range as described above were also used for PLS. Table 2 summarizes the overall PLS results including the snapshots with the best accuracy for the concentration determination of each component, the corresponding number of factors used for PLS, the standard error of calibration (SEC), and the standard error of prediction (SEP). In addition, the accuracy acquired from the room temperature measurement was also compared (designated as SEC_{RT} and SEP_{RT}). The SEC_{RT} and SEP_{RT} obtained in this study are similar to those achieved in a previous publication.[24]

Fig. 10 shows the variations in the SEPs according to the snapshots (sample measurement temperatures) for the determination of *n*-hexane (a), *n*-heptane (b), *n*-octane (c), and *n*-nonane (d) concentrations. The dashed line in each plot indicates the corresponding SEP_{RT}. The trends in the SEP-variations are quite similar to each other and the accuracies are analogously improved in the range between the 4th and 7th snapshots, which is equivalent to the approximate temperature range between −175 and −155 °C. As shown in Fig. 5, the enhancement of the spectral selectivity of these *n*-paraffins is quite substantial in the earlier stages and, therefore, it results in

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Table 2 The overall PLS results including the snapshots with the best accuracy for each component, and the corresponding number of factors used for the PLS, SEC and SEP. SEC_{RT} and SEP_{RT} indicate the errors acquired at the room temperature measurement

Component	Optimal snapshot	Number of factors	SEC (wt%)	SEP (wt%)	SEC_{RT} (wt%)	SEP_{RT} (wt%)
n-Hexane	4	12	0.43	0.62	0.56	0.96
n-Heptane	7	11	0.52	0.71	0.57	0.92
n-Octane	7	11	0.48	0.50	0.66	0.89
n-Nonane	5	12	0.46	0.67	0.59	0.93
Isooctane	10	12	0.20	0.34	0.21	0.44
Cyclohexane	7	5	0.24	0.27	0.26	0.31
Methylcyclohexane	7	10	0.12	0.19	0.13	0.20
Benzene	25	6	0.22	0.21	0.25	0.27
Toluene	37	8	0.24	0.32	0.26	0.36
Xylene	48	7	0.22	0.22	0.23	0.29
Indan	7	11	0.20	0.30	0.19	0.29

a more accurate determination of these concentrations. From approximately the 20th to the 50th snapshot, the SEP is worse than the corresponding SEP_{RT} and then they return to similar levels around the 50th snapshot. After the 50th snapshot, there is no significant variation in the SEP until the final snapshot.

The samples start thawing roughly around the 20th snapshot and turn into a liquid around the 50th snapshot as observable from the pictures in Fig. 1. So, in this period, both solid and melted liquid hydrocarbons are simultaneously present. The spectral shapes and intensities of individual components when these are either in a solid or liquid state are different as discussed earlier; therefore, Raman spectral information acquired during the bi-phasic period has higher uncertainty in representing the composition of components quantitatively. This explains the increased error during the 20th–50th snapshot range.

Meanwhile, the determination of isooctane concentration was most accurate when the spectra collected in the 10th snapshot (at approximately −140 °C) were used. The enhanced spectral selectivity of isooctane at a frozen condition, as expected from the spectral features shown in the 1230–990 cm^{-1} and 990–725 cm^{-1} ranges (refer to Fig. 5), could be the source of the improved accuracy.

In the case of naphthenic compounds (cyclohexane and methylcyclohexane), there was no significant improvement in the accuracy by measuring the samples in the low temperature range. This result was expected, because the spectral features of the samples collected in the 1st snapshot and at room temperature are not significantly different, as shown in Fig. 6, due to the presence of rigid saturated rings in their structures. These rings are relatively hard to deform by temperature perturbation.

Similar results were observed for aromatic compounds. For benzene, toluene, and xylene, the resulting accuracies were slightly improved in the 25th (approximately −111 °C), 37th (approximately −85 °C) and 48th snapshots (approximately −51 °C), respectively. In comparison to two other hydrocarbon types, the later snapshots provide the improved accuracy for these aromatic compounds. This result is possibly due to the fact that the recognition of aromatic spectral features is easier in the later snapshots when the enhanced features of the paraffin components become relatively calm. Since the aromatic spectral feature is considerably distinct due to the presence of benzene rings as discussed in a previous publication,[24] these features could be more recognizable when the bands from other components are relatively less dominant in a given spectral range.

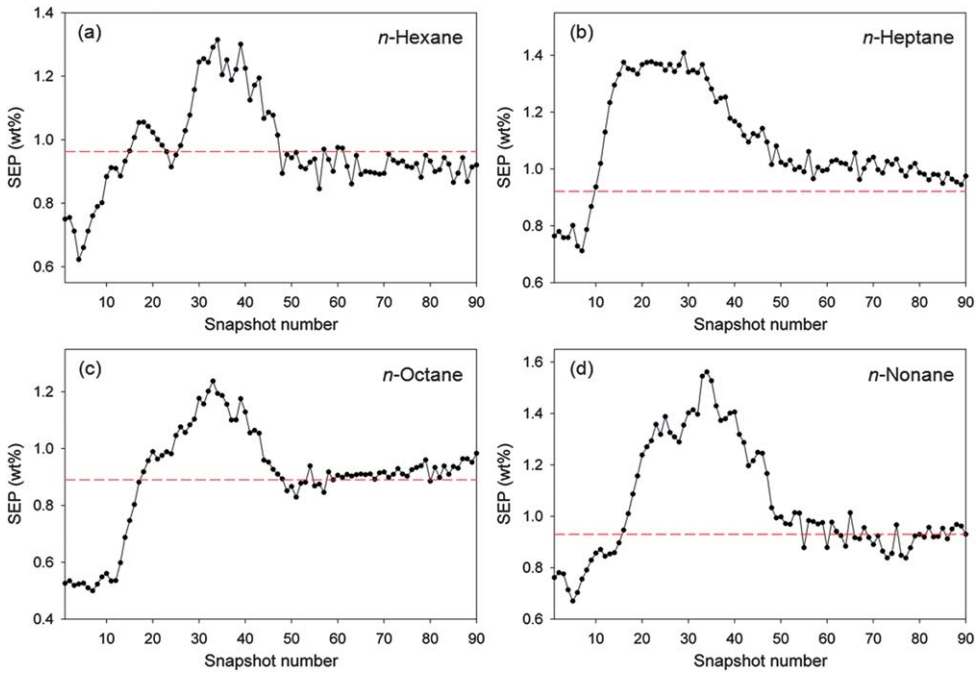

Fig. 10 Variations in the SEPs according to the snapshot number for the determination of *n*-hexane (a), *n*-heptane (b), *n*-octane (c), and *n*-nonane (d) concentrations. The dashed line in each plot indicates the corresponding SEP_{RT}.

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Published on 07 January 2013 on http://pubs.rsc.org | doi:10.1039/C2AN36575H

Conclusions

The Raman spectral collection of hydrocarbon mixture samples in low temperatures was an effective method for enhancing the spectral selectivity among the mixtures and it eventually led to improved accuracy for compositional analysis. The components with high structural deformability in a frozen state, such as linear paraffins, were primarily responsible for the enhanced selectivity. The components in the synthetic mixtures were small molecules with carbon numbers only up to 9. The strategy demonstrated in this study could be more useful for the analysis of samples composed of larger molecules with longer backbones and/or side chains, such as diesel, lube oils, oleochemicals and edible oils, which can easily deform their molecular structures in a frozen state. Ultimately, this strategy is required to evaluate for the analysis of real samples to demonstrate its practical analytical applicability. For this purpose, it will be directly utilized in order to determine the property of lube based oil samples, complex mixtures of long-chain hydrocarbons, and the possible improvement of its accuracy will be evaluated in part II of this study.[25]

Acknowledgements

This research was supported by a Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education, Science and Technology (2012R1A1B3003965).

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